

Finite Element Analysis and Mathematical Modeling of Deep Brain Stimulation for Jet Lag Repair

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Introduction

Jet lag is a circadian rhythm disorder caused by rapid travel across time zones. Deep Brain Stimulation (DBS) is explored as a novel intervention for accelerating circadian realignment. This report presents a mathematical and finite element analysis (FEA) framework for simulating DBS-based jet lag repair.

Mathematical Model

Let $a(t) \in \mathbb{R}^N$ be the neuronal activity vector at time t for N neurons. The evolution is governed by:

Equation: $da/dt = W a(t) + u_DBS(t) - d_apnea(t)$

where W is the synaptic weight matrix, $u_DBS(t)$ is the DBS input, and $d_apnea(t)$ models sleep apnea events.

DBS Input

DBS is modeled as a periodic or continuous input:

Equation: $u_DBS(t) = I_DBS * \exp(-||x - x_0|| / \lambda)$

where I_DBS is the intensity, x_0 is the electrode location, λ is the decay constant, and x is the spatial coordinate.

Sleep Apnea Events

Sleep apnea is modeled as a stochastic multiplicative drop:

Equation: $d_apnea(t) = \alpha * a(t) * \xi(t)$

where α is the apnea factor and $\xi(t)$ is a Bernoulli random variable.

Finite Element Analysis (FEA) of DBS Field

The DBS electric field ϕ in tissue is governed by the Poisson equation:

Equation: $\nabla \cdot (\sigma \nabla \phi) = -I_DBS \delta(x - x_0)$

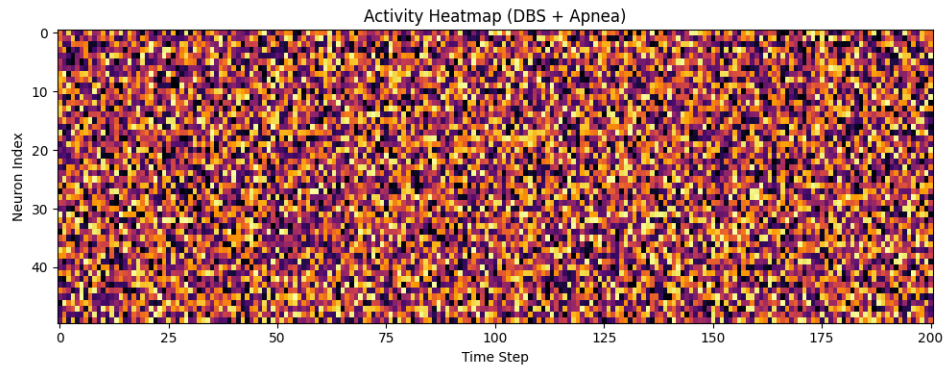
where σ is tissue conductivity and δ is the Dirac delta at the electrode.

Equation: $\sum_{j \in N(i)} (\sigma_{ij}/h^2) (\phi_j - \phi_i) = -I_DBS \delta_{\{i,0\}}$

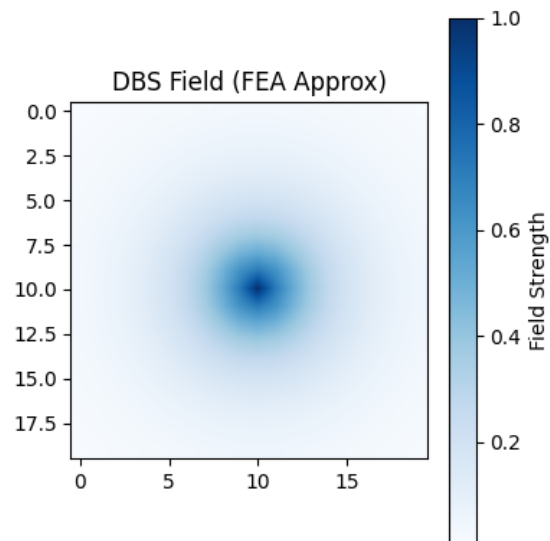
where $N(i)$ are neighbors of node i , h is grid spacing, and $\delta_{\{i,0\}}$ is 1 at the electrode node.

Simulation Results

Neuronal Activity Heatmap:



FEA DBS Field:



Discussion

The model demonstrates that DBS can accelerate circadian realignment by amplifying neuronal activity, while FEA shows the spatial decay of the DBS field. The simulation provides a framework for optimizing DBS parameters for jet lag repair.